

THE ULTIMATE KNITTING LIBRARY

The Knitter's Toolkit

Your Complete Quick-Reference Guide

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SECTION 1

Needle Size Chart

Every needle size at a glance — US, UK/Canadian, and Metric — with the yarn weight each one pairs with best.

Keep this chart next to your pattern. A quick cross-reference eliminates guesswork when switching between sizing systems.

US SIZE	METRIC (MM)	UK/CANADIAN	RECOMMENDED YARN WEIGHT
0	2.0	14	Lace / Fingering
1	2.25	13	Fingering / Super Fine
1.5	2.5	12	Fingering / Super Fine
2	2.75	12	Fingering / Super Fine
2.5	3.0	11	Sport / Fine
3	3.25	10	Sport / Fine / DK
4	3.5	9	DK / Light Worsted
5	3.75	9	DK / Light Worsted
6	4.0	8	DK / Worsted
7	4.5	7	Worsted / Aran
8	5.0	6	Worsted / Aran
9	5.5	5	Worsted / Aran
10	6.0	4	Bulky
10.5	6.5	3	Bulky
10.75	7.0	2	Bulky
11	8.0	0	Bulky / Super Bulky
13	9.0	00	Super Bulky

US SIZE	METRIC (MM)	UK/CANADIAN	RECOMMENDED YARN WEIGHT
15	10.0	000	Super Bulky
17	12.0	—	Super Bulky
19	15.0	—	Super Bulky / Jumbo
35	19.0	—	Jumbo
50	25.0	—	Jumbo

Tip

When a pattern says “4mm needles,” that is the metric size. Look across the row to find the US or UK equivalent.

Watch Out

UK/Canadian sizing runs in the opposite direction from US sizing — a higher UK number means a *smaller* needle. Always double-check with the millimeter column.

Needle Types at a Glance

Straight Needles

Classic single-pointed needles, best for flat knitting. Available in lengths from 9" to 14". Ideal for scarves, blanket panels, and dishcloths.

Circular Needles

Two short tips connected by a flexible cable. Used for knitting in the round (hats, sleeves) and for flat pieces too wide for straights. Cable lengths: 16" to 60".

Double-Pointed (DPNs)

Sets of 4 or 5 short needles pointed at both ends. Essential for small-circumference knitting: socks, mittens, and glove fingers.

Interchangeable Sets

Detachable tips that screw onto cables of varying lengths. One set replaces dozens of fixed circulars. Best investment for frequent knitters.

SECTION 2

Yarn Weight Guide

The eight standard yarn weight categories — what each one looks like, how it knits, and what to make with it.

Yarn weight is the single most important factor in choosing yarn for a project. This table covers all CYC (Craft Yarn Council) standard categories.

WEIGHT	CATEGORY	WPI	GAUGE (4"/10CM)	NEEDLE (US)	NEEDLE (MM)	COMMON USES
0	Lace	30–40+	32–34 sts	000–1	1.5–2.25	Lace shawls, doilies, wedding veils
1	Fingering / Super Fine	14–30	27–32 sts	1–3	2.25–3.25	Socks, lightweight shawls, baby clothes
2	Sport / Fine	12–18	23–26 sts	3–5	3.25–3.75	Baby items, light sweaters, wraps
3	DK / Light Worsted	11–15	21–24 sts	5–7	3.75–4.5	Sweaters, baby blankets, hats
4	Worsted / Medium	9–12	16–20 sts	7–9	4.5–5.5	Sweaters, blankets, scarves, amigurumi
5	Bulky / Chunky	7–9	12–15 sts	9–11	5.5–8.0	Thick hats, cowls, rugs, quick blankets
6	Super Bulky / Roving	5–7	7–11 sts	11–17	8.0–12.0	Quick throws, bulky cowls, arm-knit projects
7	Jumbo	1–5	6 sts or fewer	17+	12.0–25.0	Arm knitting, giant blankets, extreme knits

What Is WPI?

Wraps Per Inch — wrap the yarn around a ruler for one inch without stretching or leaving gaps. Count the wraps. This is the fastest way to identify an unlabeled yarn’s weight.

Quick Fiber Reference

Animal Fibers

Wool — warm, elastic, felts easily. **Merino** — soft, non-itchy wool. **Alpaca** — silky warmth, no memory. **Cashmere** — luxury softness, delicate. **Mohair** — fluffy halo, strong fiber. **Silk** — beautiful drape, cool hand.

Plant & Synthetic Fibers

Cotton — crisp, cool, no stretch. **Linen** — sturdy, softens with washing. **Bamboo** — silky drape, breathable. **Acrylic** — budget-friendly, machine washable. **Nylon** — adds strength (great in sock blends). **Polyester** — durable, quick-dry.

SECTION 3

Gauge Calculator

How to make a swatch, measure it accurately, and calculate the adjustments that make your finished piece fit perfectly.

Gauge is the number of stitches and rows per unit of measurement. Skipping gauge is the number-one reason projects end up the wrong size.

How to Make a Gauge Swatch

- 1 **Cast on extra stitches.** If the pattern says 20 sts = 4", cast on at least 28–32 stitches. You need a border of several stitches on each side — edge stitches distort the measurement.
- 2 **Knit at least 5 inches.** Work in the stitch pattern specified (usually stockinette). Knit enough rows so you can measure 4" of fabric away from the cast-on and bind-off edges.
- 3 **Bind off loosely and wash.** Treat the swatch exactly as you will treat the finished item — hand wash, machine wash, or block it. Yarn can change dramatically after washing.
- 4 **Lay flat and let dry completely.** Pin it out if the pattern calls for blocking. Do not stretch it.

How to Measure

- 1 **Place a ruler horizontally** across the center of the swatch (away from edges). Count the number of stitches across exactly 4 inches (10 cm). Include half-stitches.
- 2 **Place the ruler vertically** and count the number of rows in 4 inches.
- 3 **Repeat at least twice** in different spots. Average the results for accuracy.

The Gauge Formula

Stitch Gauge Formula

Your gauge = stitches counted ÷ inches measured

Stitches needed = desired width (inches) × your stitch gauge

Rows needed = desired length (inches) × your row gauge

Worked Example

Pattern gauge: 20 sts and 28 rows = 4" on US 7 (4.5mm) needles.

Your swatch: You count 22 sts and 30 rows in 4".

Your stitch gauge is tighter. You have two options:

- 1 Change needle size (preferred).** Go up one needle size (US 8 / 5.0mm) and swatch again. Larger needles produce fewer stitches per inch. Repeat until you match the pattern gauge.
- 2 Recalculate stitch counts.** If the pattern says to cast on 100 sts for a 20" chest panel: $20" \times (22 \text{ sts} \div 4") = 20 \times 5.5 = 110 \text{ sts}$. You would need 110 stitches instead of 100.

Adjustment Quick-Reference

PROBLEM	YOUR GAUGE VS. PATTERN	FIX
Too many stitches per inch	Tighter	Go up 1 needle size
Too few stitches per inch	Looser	Go down 1 needle size
Stitch gauge matches, row gauge off	Mixed	Adjust length by working more or fewer rows; measure in inches rather than counting rows
Both off by more than 1 st per inch	Very different	Try a completely different needle material (wood vs. metal) or re-evaluate yarn weight

Common Mistake

Measuring the swatch while it is still on the needles or before washing. Always bind off, wash, dry, and then measure.

Gauge Swatch Checklist

- ☐ Cast on at least 8 extra stitches beyond the gauge measurement
- ☐ Worked in the correct stitch pattern
- ☐ At least 5" (13 cm) tall
- ☐ Bound off and removed from needles
- ☐ Washed and dried the same way the finished piece will be
- ☐ Measured in multiple spots and averaged
- ☐ Stitch gauge recorded: ____ sts / 4"
- ☐ Row gauge recorded: ____ rows / 4"

SECTION 4

Abbreviation Glossary

Over 100 standard knitting abbreviations so you can read any pattern with confidence.

Knitting patterns use abbreviations to save space. This glossary covers the ones you will encounter in virtually every pattern, plus many advanced ones.

ABBR.	MEANING
alt	alternate
approx	approximately
beg	begin / beginning
bet	between
BO	bind off
BOR	beginning of round
C4B	cable 4 back: slip 2 sts to cn, hold in back, k2, k2 from cn
C4F	cable 4 front: slip 2 sts to cn, hold in front, k2, k2 from cn
CC	contrast color
cn	cable needle
CO	cast on
cont	continue
CDD	central double decrease: sl 2 sts tog knitwise, k1, pass slipped sts over
dec	decrease / decreasing
dpn(s)	double-pointed needle(s)
EOR	every other row
est	established

ABBR.	MEANING
foll	following
g st	garter stitch (knit every row)
inc	increase / increasing
k	knit
k1b / k1-b	knit 1 through the back loop
k2tog	knit 2 stitches together (right-leaning decrease)
k3tog	knit 3 stitches together
kfb	knit into front and back of same stitch (increase)
kw	knitwise
LH	left-hand
lp(s)	loop(s)
m	marker
M1	make 1 stitch (generic lifted increase)
M1L	make 1 left: lift bar between sts from front to back, knit through back loop
M1R	make 1 right: lift bar between sts from back to front, knit through front loop
M1P	make 1 purlwise
MC	main color
meas	measures
mult	multiple
oz	ounce(s)
p	purl
p2tog	purl 2 stitches together
pat(s) / patt	pattern(s)

ABBR.	MEANING
pfb	purl into front and back of same stitch (increase)
pm	place marker
prev	previous
pssso	pass slipped stitch over
p2sso	pass 2 slipped stitches over
pu	pick up
pw	purlwise
rem	remain / remaining
rep	repeat
rev St st	reverse stockinette stitch (purl side is RS)
RH	right-hand
rib	ribbing
rnd(s)	round(s)
RS	right side (public side of fabric)
sc	single crochet (used in some finishing)
sk	skip
sk2p	slip 1, knit 2 together, pass slipped stitch over (double decrease)
SKP / skpo	slip, knit, pass slipped stitch over (same as ssk in some patterns)
sl	slip
sl st	slip stitch
sl1k	slip 1 knitwise
sl1p	slip 1 purlwise
sm	slip marker

ABBR.	MEANING
ssk	slip, slip, knit (left-leaning decrease)
sssk	slip, slip, slip, knit (double decrease)
ssp	slip, slip, purl through back loops
st(s)	stitch(es)
St st	stockinette stitch (knit RS, purl WS)
tbl	through the back loop
tog	together
w&t	wrap and turn (short rows)
WS	wrong side (back side of fabric)
wyib	with yarn in back
wyif	with yarn in front
yd(s)	yard(s)
yo	yarn over (wrap yarn around needle to create a new stitch / eyelet)
yo2	yarn over twice (double yarn over)
yrn	yarn round needle (UK term for yo between purl stitches)

Symbols You May See in Charts

SYMBOL	MEANING (RS)	MEANING (WS)
⬜ (empty box)	Knit	Purl
•	Purl	Knit
O	Yarn over	Yarn over
/ (right slant)	K2tog	P2tog
\ (left slant)	SSK	SSP
><	Cable cross	—

SYMBOL	MEANING (RS)	MEANING (WS)
■ (filled box)	No stitch (placeholder)	No stitch
V	Knit into front & back (kfb)	Pfb
Λ	Central double decrease (CDD)	—

Reading Tip

Charts are read right to left on RS rows and left to right on WS rows — just like the direction you are actually knitting. Odd-numbered rows are usually RS.

SECTION 5

Conversion Tables

Yards to meters, ounces to grams, inches to centimeters — every conversion you need while working from patterns in different measurement systems.

Many patterns originate in countries that use metric, while yarn labels in North America show imperial. These tables make switching painless.

Length: Yards ↔ Meters

YARDS	METERS	YARDS	METERS
1	0.91	150	137
5	4.57	200	183
10	9.14	250	229
25	22.9	300	274
50	45.7	400	366
75	68.6	500	457
100	91.4	1000	914

Quick Formula

Yards × 0.9144 = Meters | Meters × 1.0936 = Yards

Weight: Ounces ↔ Grams

OUNCES	GRAMS	OUNCES	GRAMS
0.5	14	4	113
1	28	5	142
1.75	50	6	170
2	57	7	198

OUNCES	GRAMS	OUNCES	GRAMS
3	85	8	227
3.5	100	16 (1 lb)	454

Quick Formula

Ounces × 28.3495 = Grams | Grams × 0.03527 = Ounces

Length: Inches ↔ Centimeters

INCHES	CENTIMETERS	INCHES	CENTIMETERS
0.25	0.64	12	30.5
0.5	1.27	14	35.6
1	2.54	16	40.6
2	5.08	18	45.7
4	10.2	20	50.8
6	15.2	24	61.0
8	20.3	36	91.4
10	25.4	48	121.9

Quick Formula

Inches × 2.54 = Centimeters | Centimeters × 0.3937 = Inches

Common Skein Sizes

YARN WEIGHT	TYPICAL SKEIN	APPROX. YARDAGE	APPROX. METERS
Lace	50 g / 1.75 oz	400–600 yds	366–549 m
Fingering	50 g / 1.75 oz	200–250 yds	183–229 m
Sport	50 g / 1.75 oz	140–170 yds	128–155 m
DK	50 g / 1.75 oz	120–150 yds	110–137 m

YARN WEIGHT	TYPICAL SKEIN	APPROX. YARDAGE	APPROX. METERS
Worsted	100 g / 3.5 oz	200–230 yds	183–210 m
Bulky	100 g / 3.5 oz	100–140 yds	91–128 m
Super Bulky	100 g / 3.5 oz	60–90 yds	55–82 m
Jumbo	250 g / 8.8 oz	40–70 yds	37–64 m



SECTION 6

Care Symbols Guide

Decode the symbols on yarn labels so your finished pieces stay beautiful wash after wash.

Standard textile care symbols are used worldwide. They appear on yarn bands and tell you exactly how to wash, dry, iron, and clean your knitted items.

Washing

Tub icon = machine washable. The number inside indicates maximum temperature in °C.

30 °C — Cold / delicate wash

40 °C — Warm wash

60 °C — Hot wash

Tub with hand — Hand wash only (max 40 °C)

Tub with X — Do not wash (dry clean only)

Bar under tub — Gentle / permanent press cycle

Double bar — Extra gentle / delicate cycle

Bleaching

Triangle = bleaching allowed.

Empty triangle — Any bleach allowed

Triangle with diagonal lines — Non-chlorine / oxygen bleach only

Triangle with X — Do not bleach

For knitters: Most animal fibers (wool, alpaca, silk) must never be bleached. Cotton and linen tolerate oxygen bleach. When in doubt, skip the bleach.

Drying

Square icon = drying instructions.

Square with circle — Tumble dry allowed

One dot in circle — Low heat

Two dots in circle — Normal / medium heat

Square+circle with X — Do not tumble dry

Square with horizontal line — Dry flat
(standard for most knits)

Square with arc at top — Drip dry / line dry

Ironing

Iron icon = ironing instructions.

One dot — Low heat (110 °C / 230 °F) —
synthetics

Two dots — Medium heat (150 °C / 300 °F) —
wool, silk

Three dots — High heat (200 °C / 390 °F) —
cotton, linen

Iron with X — Do not iron

Iron with X under — Do not steam

Tip: Most knits should be steamed or blocked
rather than pressed directly with an iron.

Dry Cleaning

Circle icon = professional cleaning.

P in circle — Dry clean with
perchloroethylene (standard solvent)

F in circle — Dry clean with petroleum solvent
only

W in circle — Professional wet clean

Circle with X — Do not dry clean

Bar under circle — Gentle process

Fiber Care Quick Reference

Wool / Alpaca: Hand wash cold, lay flat to
dry, no bleach

Merino Superwash: Machine wash gentle
cold, lay flat to dry

Cotton / Linen: Machine wash warm, can
tumble dry low

Silk: Hand wash cold or dry clean, do not
wring

Acrylic: Machine wash warm, tumble dry low

Bamboo: Hand wash or gentle cycle, lay flat

Cashmere: Hand wash cold, lay flat, no
wringing

Important

Always check the yarn label before washing a finished item. Different dye lots of the same yarn can sometimes have different care instructions.

SECTION 7

Yarn Substitution Guide

How to confidently swap one yarn for another without ruining your project.

The yarn a pattern calls for might be discontinued, too expensive, or unavailable in your area. A smart substitution keeps your project on track.

The Substitution Decision Process

- 1 Match the yarn weight category.** This is the most critical factor. A DK substitute for a DK yarn is far more likely to work than crossing weight classes.
- 2 Check the gauge.** Even within the same weight category, gauge can vary. Swatch with the new yarn on the recommended needle size and compare to the pattern gauge.
- 3 Consider the fiber content.** Fiber affects drape, warmth, stretch, and stitch definition. A cotton DK and a wool DK behave very differently in a cable pattern.
- 4 Calculate the total yardage needed.** Substitute by yardage, not by number of skeins. If the pattern calls for 5 skeins at 220 yds each (1,100 yds total), you need 1,100 yds of the new yarn, regardless of its skein size.
- 5 Buy an extra skein.** Different yarn structures use fiber at different rates. An extra skein is cheap insurance.
- 6 Knit a full swatch.** Wash and block it. Check gauge, drape, and how you feel about the fabric before committing.

Yardage Calculation

Formula

Total yards needed = (pattern skeins) × (yards per skein of original yarn)

Skeins of new yarn = total yards needed ÷ (yards per skein of new yarn)

Always round UP and add one extra skein.

Fiber Substitution Considerations

IF THE PATTERN USES	GOOD SUBSTITUTES	CAUTION WITH
Wool	Merino, wool blends, alpaca (for warmth)	Cotton (no memory/stretch), acrylic (different hand)
Cotton	Linen, bamboo, cotton blends	Wool (different drape, much warmer), stretchy yarns
Alpaca	Wool (for structure), alpaca blends, cashmere	Cotton (no warmth), plant fibers (no loft)
Acrylic	Acrylic blends, superwash wool (similar care)	Delicate fibers requiring hand wash
Silk	Bamboo, Tencel, silk blends	Wool (less drape), cotton (less sheen)
Linen	Cotton, hemp, bamboo	Wool or acrylic (completely different character)

When Fiber Properties Matter Most

Stitch Definition Critical Cables, textured patterns, colorwork. Use smooth, plied yarn with good memory (wool, wool blends). Avoid fuzzy singles or high-halo yarns.	Drape Critical Shawls, flowing garments, lace. Use fibers with drape: silk, bamboo, linen, Tencel. Avoid springy, bouncy wool unless the pattern was designed for it.
Warmth Critical Winter accessories, heavy sweaters. Use wool, alpaca, cashmere, or warm blends. Plant fibers and silk provide little insulation.	Durability Critical Socks, bags, high-wear items. Use superwash wool/nylon blends, cotton, or acrylic. Avoid pure alpaca or silk (pill or wear quickly at stress points).

Common Substitution Pairs

ORIGINAL YARN (EXAMPLE)	RELIABLE SUBSTITUTE OPTIONS	WEIGHT
100% Merino worsted	Any superwash merino worsted; merino/nylon blend worsted	4 — Worsted

ORIGINAL YARN (EXAMPLE)	RELIABLE SUBSTITUTE OPTIONS	WEIGHT
Alpaca DK	Alpaca/wool blend DK; merino DK (add warmth layer)	3 — DK
Cotton sport	Cotton/acrylic blend sport; bamboo sport	2 — Sport
Sock yarn (fingering, wool/nylon)	Any fingering-weight wool/nylon blend; superwash merino fingering	1 — Fingering
Bulky single-ply wool	Bulky plied wool; bulky wool/acrylic blend	5 — Bulky
Silk lace	Silk/merino lace blend; Tencel lace; bamboo lace	0 — Lace

Substitution Checklist

- ☐ Same yarn weight category
- ☐ Gauge swatch matches pattern gauge (after washing)
- ☐ Similar fiber content or compatible drape/stretch
- ☐ Total yardage calculated (not just skein count)
- ☐ Extra skein purchased
- ☐ Care requirements are acceptable
- ☐ Fabric hand and appearance are satisfying

Don't Skip the Swatch

A substitution that looks perfect on paper can still fail in practice. Always knit, wash, and block a swatch before buying all the yarn for a large project.

Next Step

Keep this toolkit within reach every time you knit. Open **The Great Stitch Library** to pick your next stitch pattern, then use the needle chart, gauge guide, and abbreviation glossary here to follow the instructions with confidence.